

PLASMA CUTTING

The Complete Technical Poster — Principle • Process • Parameters • Quality • Safety • Applications

1. What is plasma cutting?

Plasma cutting is a **thermal cutting process** that severs electrically conductive metals (steel, stainless steel, aluminium, copper, brass, titanium, cast iron...) using a **high-velocity jet of ionised gas — plasma** — heated by an electric arc to extreme temperature. The jet melts the metal along the cut line while the gas blast blows the molten material out of the kerf (the slot left behind). It works on any material that conducts electricity, which makes it far more versatile than oxy-fuel cutting.

2. The physics — why it works

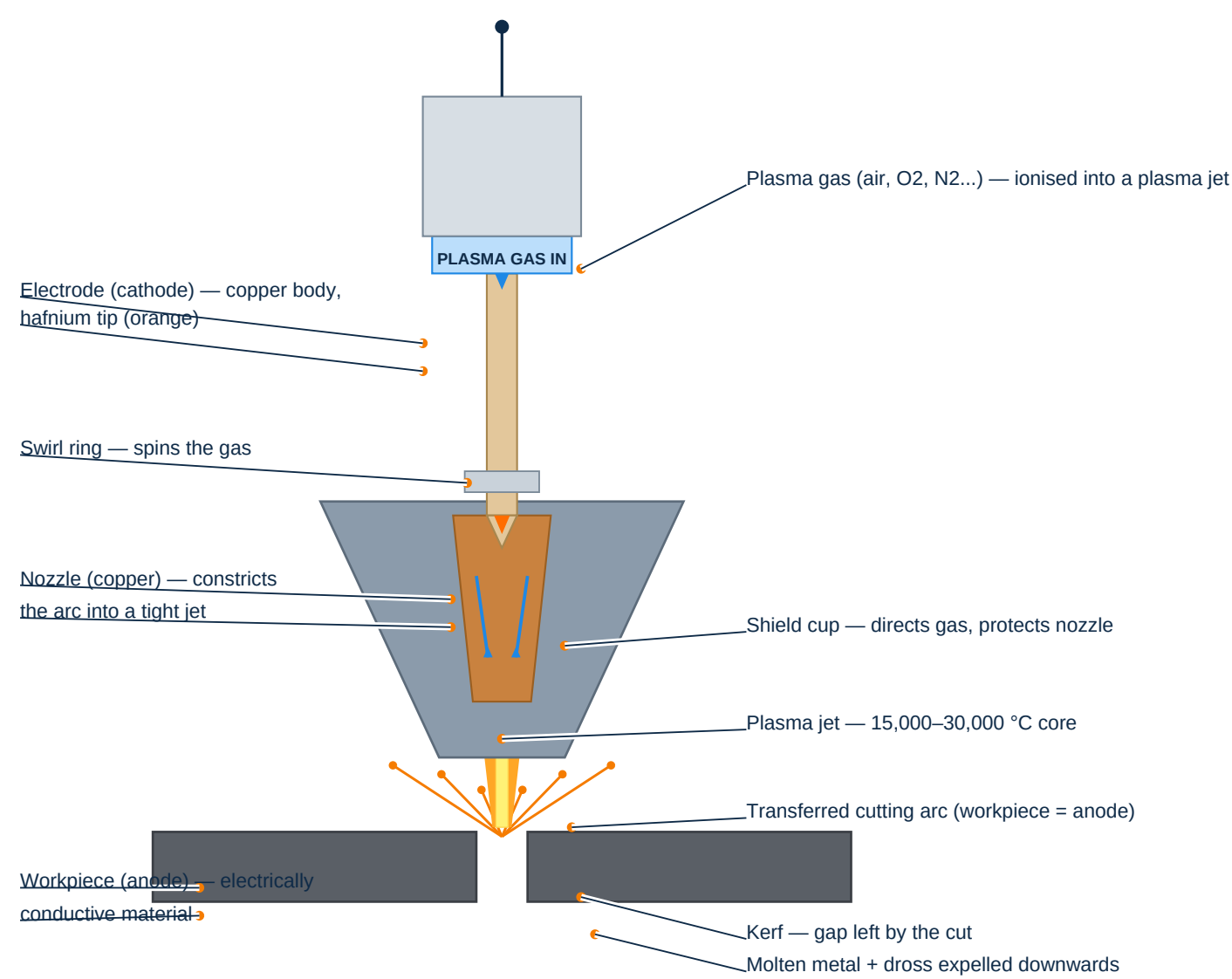
Plasma is the 4th state of matter. When a gas is heated enough, its atoms gain so much energy that electrons are stripped away and it becomes a mixture of free electrons and positive ions — electrically conductive and able to carry an arc.

Ionisation: a high-voltage spark (pilot arc) creates a conductive path of ionised gas between the torch electrode and the nozzle tip.	Arc transfer: the arc then jumps to the workpiece (transferred arc), where nearly all the heat is delivered to the metal instead of the torch.	Extreme temperature: the plasma core reaches 15,000–30,000 °C — several times hotter than the surface of the Sun (5,500 °C).
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3. How a cut happens — step by step

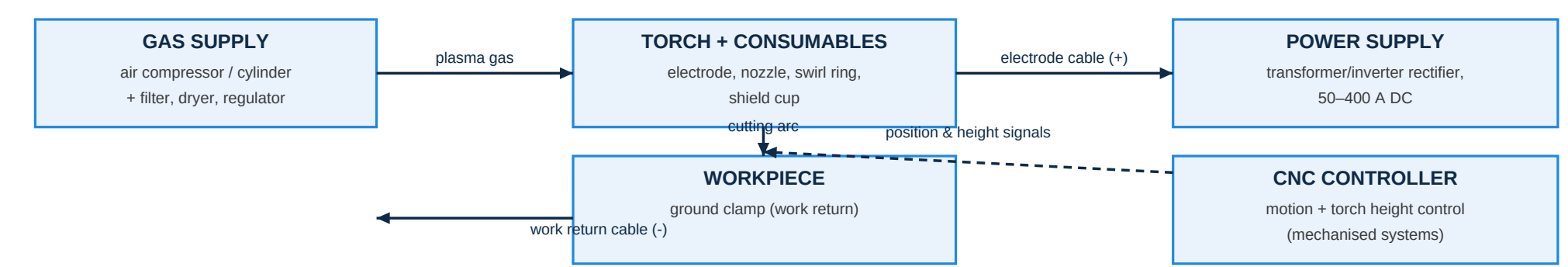
- Gas flows:** compressed gas (shop air, oxygen, nitrogen...) enters the torch and is spun by the swirl ring.
- Pilot arc starts:** a high-frequency, high-voltage spark ionises the gas between the electrode and nozzle — this small arc is called the **pilot (non-transferred) arc**.
- Arc transfers:** as the torch approaches the workpiece, the arc jumps to the metal — now called the **transferred arc**. The workpiece becomes the anode; virtually all heat goes into the cut.
- Gas becomes plasma:** the arc superheats the constricted gas into a plasma jet that exits the nozzle at up to **3–6 times the speed of sound** (supersonic).
- Melting:** the jet's enormous heat melts the metal instantly along the cut line.
- Expulsion:** the high-velocity gas blows the molten metal out through the bottom of the cut, forming the kerf.
- Motion:** the torch moves along the programmed path (hand-guided or CNC) and the cut follows. **Shielding:** shield gas around the jet protects the cut edge and cools the nozzle.

4. Anatomy of a plasma torch



Part	Material / role	Why it matters
Electrode (cathode)	Copper body with a small hafnium insert	Hafnium withstands the extreme arc temperature; it erodes with every arc start — the main consumable to replace.
Nozzle	Copper with a precision orifice	Constricts the arc into a tight jet. A worn or damaged orifice ruins cut quality (bevel, dross, one-sided cut).
Swirl ring	Insulating ring with angled gas ports	Spins the plasma gas — the swirl stabilises the arc and keeps it centred.
Shield cup / retaining cap	Stainless / ceramic	Directs secondary shielding gas over the cut edge and protects the nozzle from spatter.
Torch body & leads	Insulated handle, electrode cable	Carries current + gas to the head; must be kept clean and dry.

5. The complete cutting system



System part	Function	Key detail
Power supply	Converts mains AC to DC cutting current (typically 20–400 A)	Inverter units are light and efficient; output current sets cut speed and max thickness.
Gas supply	Provides plasma gas (and shielding gas)	Must be clean, dry and oil-free — moisture is the #1 killer of consumable life.
Torch & consumables	Creates and directs the plasma jet	Drag (hand) tips or machine torches; consumable set as in section 4.
Work return (ground) clamp	Completes the electric circuit to the workpiece	A poor ground causes arc instability and wasted consumables.
CNC table (mechanised)	Moves the torch on X/Y axes at programmed speed; holds height	Torch height control (THC) senses arc voltage and keeps standoff constant over warped plate.
Fume extraction / water table	Removes fume, catches sparks and dross	Water tables also reduce noise, fumes and sheet distortion.

6. Plasma gases — choosing the right one

Gas	Typical use	Notes
Compressed shop air	General purpose; mild steel and most metals	Cheapest option. Must be dry and oil-free; shortens consumable life vs pure gases.
Oxygen (O2)	Mild / carbon steel (mechanised)	Fastest cuts, cleaner edges, less dross — but burns the hafnium electrode faster.
Nitrogen (N2)	Stainless steel and aluminium	Bright, oxide-free edges; standard for mechanised cutting of these metals.
Argon–hydrogen (Ar–H2, ~65/35)	Thick stainless & aluminium (12 mm+)	Highest energy and hottest jet; smooth edges; expensive.
F5 (95% N2 / 5% H2)	Mechanised stainless / aluminium	Good edge quality at lower cost than Ar–H2.

7. Materials — what plasma can (and cannot) cut

Material	Cutability	Notes
Mild / carbon steel	Excellent	The most common material; air or O2; clean edges, low dross when parameters are right.
Stainless steel	Good	Use N2 / F5 / Ar–H2; air leaves a dark oxide edge that may need grinding.
Aluminium	Good	Use N2 (or Ar–H2 for thick plate); manage dross; oxide film is removed by the arc.
Copper & brass	Fair	High thermal conductivity needs high current; edges rougher; thick sections struggle.
Cast iron	Fair	Cuttable without preheat; can harden the cut edge (machinability issue).
Titanium	Good (dry gas)	Use dry N2; must shield the hot edge from air contamination.
Galvanised steel	Good	Zinc coating produces toxic fumes — extraction is mandatory.
Non-conductors (wood, plastic, glass, stone)	NO	Plasma needs an electrical circuit through the workpiece — non-conductive materials cannot be cut.

8. Cutting parameters — typical values (air plasma, mild steel)

Numbers below are **indicative** — always use the manufacturer's **cut chart** for your exact torch, consumables and gas. Amperage, speed, standoff and gas pressure all interact: change one and the others may need adjusting.

Plate thickness	Current (A)	Approx. speed (mm/min)	Standoff (mm)	Nozzle / note
3 mm	30	1200–1500	1.5–3 (drag tip)	Fine nozzle; fast, clean cut
6 mm	45	800–1000	2–3	Standard nozzle
10 mm	60	500–700	3–4	Standard nozzle
16 mm	80	350–450	3–5	Large nozzle
25 mm	100	200–300	4–6	Large nozzle; may need 2 passes on some units
50 mm	200 (mechanised)	100–150	4–6 + THC	Heavy-duty; piercing needs high standoff first
Severance (max)	unit rating	slow	—	Unit's absolute max thickness — rough edge, not a quality cut

9. How each parameter affects the cut

If you increase...	Effect on the cut	Watch out for
Cutting current (amps)	Faster cutting, wider kerf, cuts thicker plate	Too much: more dross, double arcing, rapid consumable wear.
Travel speed	Less heat input, smaller HAZ, less dross	Too fast: the jet lags (drag lines), bevel increases, cut may not penetrate.
Standoff (torch height)	Wider kerf, more top-edge rounding	Too high: arc becomes unstable and the bevel worsens; too low: nozzle can touch the plate and short (double arc = destroyed nozzle).
Gas pressure	More momentum to blow dross away	Too high: rough edge, arc blow-out; too low: weak arc start, short consumable life.
Piercing (starting on the plate)	—	Always pierce with the torch high, pause to let the jet blow through, THEN lower to cut height — protects the nozzle from molten metal splash-back.

10. Cut quality — what the edge tells you

Feature	What it is	Acceptable / how to improve
Kerf	Width of material removed by the jet (nozzle orifice + ~0.5 mm)	Typically 1.5–3 mm; narrower with high-definition torches. Wider kerf = more material waste.
Bevel (edge squareness)	Angle of the cut edge from 90°; caused by the arc leaning as the jet exits	Standard plasma: 1–4°; high-definition: 1–2° or less. Reduce standoff, replace worn nozzle.
Dross	Resolidified metal stuck to the bottom (or top) edge	Bottom dross = too slow / low amps; top dross = too fast. Adjust speed, height and current.
HAZ (heat-affected zone)	Metal beside the cut whose structure changed from the heat	Smaller than oxy-fuel, bigger than laser. Less heat input (faster speed) shrinks it.
Warping / distortion	Sheet buckling from uneven heating	Use faster speed, water tables, cut in a balanced sequence, clamp thin sheet.
Drag lines	Fine curved lines on the cut face showing jet lag	Straight vertical lines = ideal speed. Deep curves = too slow.
Top-edge rounding	Melting of the top corner of the cut	Standoff too high; lower the torch.

11. Troubleshooting — common problems and fixes

Symptom	Likely cause(s)	Fix
Excessive dross on bottom edge	Speed too slow; current too low; standoff wrong	Raise speed or amps; re-check torch height.
Dross on top edge	Speed too fast	Slow the travel speed.
Excessive bevel on both sides	Standoff too high; worn nozzle; gas pressure high	Lower torch; fit new nozzle; reduce pressure.
Cut leans / cuts better on one side	Worn or half-blocked nozzle orifice; torch tilted	Replace nozzle; keep torch perpendicular to the plate.
Nozzle damaged (double arcing)	Standoff too low; long piercing at cut height; moisture in gas	Pierce high, then drop; dry the air; correct height.
Short consumable life	Moist/oily air; too many starts; amps too high	Fit and drain a dryer/filter; minimise pilot time; respect cut chart.
Arc won't start	Bad ground clamp; worn electrode; faulty pilot circuit	Clean/retighten ground; replace electrode; check torch switch and leads.
Arc blows out / wanders	Gas pressure too high; worn swirl ring; arc too long	Reduce pressure; replace swirl ring; lower standoff.
Cut not penetrating	Amps too low; speed too fast; plate beyond unit rating	Check cut chart; reduce speed; use a bigger unit.

12. Plasma vs other cutting processes

Property	Oxy-fuel	Plasma	Laser	Waterjet
Principle	Gas flame + oxygen jet burns steel	Ionised gas arc melts + blows metal	Focused light beam melts/vapourises	Abrasive water erodes material
Materials	Mild/carbon steel only	All electrically conductive metals	Most metals (thin); some non-metals	Almost anything (metal, stone, glass, composites)
Max thickness	300+ mm	~150 mm	~25–30 mm (steel)	200+ mm
Edge quality	Rough; heavy dross; large HAZ	Good; high-definition approaches laser	Excellent; square, smooth edges	Excellent; no heat at all
Kerf width	2–5 mm	1.5–3 mm	0.1–0.5 mm	0.8–1.5 mm
Heat-affected zone	Very large	Medium (small with fast speeds)	Medium–low	None (cold cutting)
Speed on 10 mm steel	Slow (~0.3 m/min)	Fast (~0.6–1 m/min)	Very fast on thin sheet, slower on thick	Slow
Equipment cost	Low	Medium	High	Very high
Best for	Very thick structural steel, demolition	General fabrication; any metal; mid thicknesses	Precision thin sheet work	Heat-sensitive and exotic materials

13. Advantages and limitations

Advantages of plasma cutting - Cuts any electrically conductive metal — mild steel, stainless, aluminium, Faster than oxy-fuel on steel up to ~50 mm thick — often 3–10× the speed. Narrow kerf and small heat-affected zone compared with oxy-fuel. - No preheat required — the jet is hot instantly, saving time and gas. Portable hand-held units from ~15 A (thick-sheet work starts around 30–60 A) - Cuts painted, rusty or stacked plate without preparation. Cheap consumables and low running cost vs laser (especially with shop air) - Easily automated — CNC plasma tables deliver repeatable, nested parts from CAD drawings.	Limitations - Only works on conductive materials — no wood, plastic, glass or stone. - Cut edges are not perfectly square (1–4° bevel) — laser and waterjet are more precise Loud (95–115 dB) — hearing protection required. - Produces fumes, UV arc flash and spatter — ventilation and PPE essential. - Consumables (electrode, nozzle) wear out and affect quality as they age. - Thick, precision plate work is better done by laser/waterjet; very thick steel by oxy-fuel - Higher initial cost than oxy-fuel equipment.
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14. Safety — plasma cutting is dangerous

Hazard	Risk	Control
Electric shock	Open-circuit voltages of 200–400 V	Insulated gloves, dry boots and floor; correct earthing; never touch the nozzle while the pilot arc is on.
Arc flash (UV / IR light)	Arc eye (welder's flash), skin sunburn	Auto-darkening welding helmet with shade 8–13 filter; full cover clothing; shield bystanders with screens.
Molten metal spatter	Burns to skin and eyes	Leather gloves, flame-resistant jacket, closed shoes; keep hands out of the spark stream.
Noise	95–115 dB — permanent hearing damage	Ear defenders or plugs at all times.
Fumes and gases	Metal-oxide fume, ozone, nitrogen oxides; zinc/chrome fume is highly toxic	Local exhaust ventilation / fume extractor at the cut; respirator in confined spaces; never cut coated metal indoors without extraction.
Fire	Sparks and hot dross ignite flammables up to ~10 m away	Clear the area, wet or cover combustibles, have a fire extinguisher and a fire watch for hot work.
Compressed gas	Cylinder and regulator failure	Secure cylinders upright; use correct regulators; never oil oxygen fittings; check hoses.
Electromagnetic interference	Disrupts nearby electronics / pacemakers	Follow earthing and bonding rules; keep leads tidy and close together.

15. Applications

Fabrication and sheet-metal shops · automotive repair (body panels, exhausts, chassis) · structural steel and construction · shipbuilding and marine repair · agricultural and earth-moving equipment · HVAC ductwork · sign-making and metal art (CNC letters, sculptures) · demolition and scrap-yard cutting · rail and heavy-plant maintenance · custom motorcycle and automotive parts. Wherever conductive metal must be cut fast and cheaply, plasma is usually the first choice.
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16. Consumables and maintenance

Item	Typical life	Signs of wear / action
Electrode (hafnium)	Hundreds of starts (300–1,000+)	Deep pit in the hafnium tip — replace. Never rotate a worn electrode.
Nozzle	Similar to electrode; they wear together	Orifice becomes oval or enlarged — poor cuts, one-sided bevel — replace.
Swirl ring	Long	Blocked gas ports — arc wanders — clean or replace.
Shield cup	Long	Spatter build-up — clean with a dedicated tool; cracked ceramic — replace.
Air filter / dryer	Service regularly	Water in the air stream ruins consumables fast — drain the separator daily.

Daily: drain water from the compressor/separator; check hoses and cables for damage; check the ground clamp. **Weekly:** inspect consumables and replace worn parts as a set; clean the torch head. **Monthly:** verify air pressure and quality at the torch; check torch O-rings and insulation; test the pilot-arc start. A clean, dry system is the cheapest way to buy consistent cut quality.

17. History — how plasma cutting developed

Year	Milestone
1940s	Arc-based cutting experiments; oxy-fuel dominates industry.
1955	Plasma arc cutting invented at Union Carbide (Linde Division) — used with argon–hydrogen for stainless steel and aluminium.
1960s	Dual-flow and water-injection torches improve edge quality and cut thick plate.
1970s	Compressed-air plasma arrives — small, affordable hand-held units make plasma cutting accessible to workshops and hobbyists.
1980s	CNC plasma tables and automatic torch-height control bring repeatable, high-speed production cutting.
1990s	High-definition (precision) plasma: tighter kerf, squarer edges, near-laser quality on plate.
2000s–today	Inverter power supplies (lighter, more efficient), fine-feature cutting for thin sheet, robotic plasma cells, and competition from fibre lasers on thin material.

18. Glossary

Term	Meaning
Plasma	Ionised gas — a mixture of free electrons and positive ions that conducts electricity.
Pilot arc	The small non-transferred arc between electrode and nozzle that ionises the gas before the main arc transfers.
Transferred arc	The main cutting arc, which passes from the torch through the workpiece (workpiece = anode).
Cathode / anode	Negative electrode (torch electrode) / positive electrode (workpiece).
Kerf	The width of material removed by the cut.
Dross	Resolidified metal clinging to the underside (or top) of the cut edge.
HAZ	Heat-affected zone — the metal beside the cut whose microstructure was changed by heat.
Bevel	The angle of the cut edge away from 90°.
Standoff	The distance between the nozzle tip and the workpiece.
Drag tip	A nozzle designed to slide directly on the plate for hand cutting.
THC	Torch height control — automatically keeps standoff constant by sensing arc voltage.
Cut chart	The manufacturer's table of current, speed, gas pressure and standoff for each material and thickness.
Severance cut	The absolute maximum thickness a unit can cut — slow, rough, and not a production-quality cut.
Water table	A cutting bed with water to catch dross, suppress fume, noise and distortion.

References: manufacturer cut charts (Hypertherm, ESAB, Lincoln Electric); process handbooks. Values are indicative — always follow the equipment manual for your specific torch.